

Human Performance Modeling - 4

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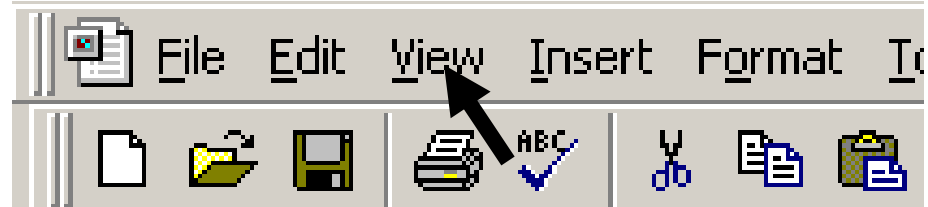
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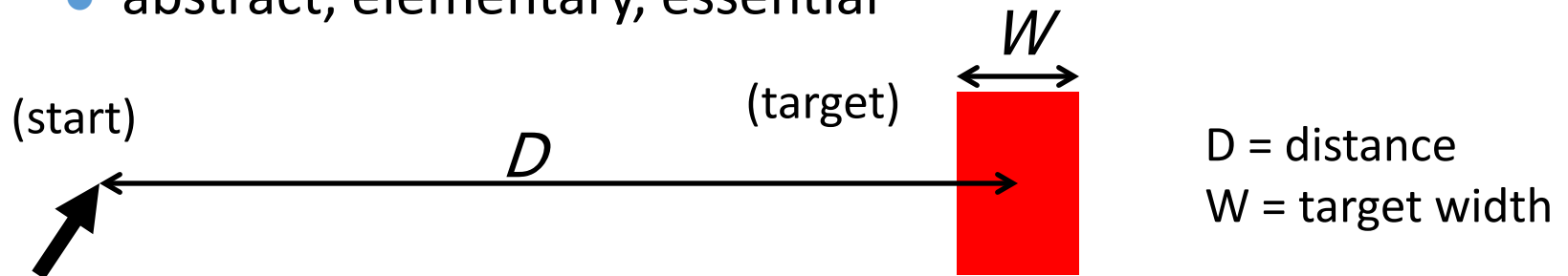
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Pointing Tasks

- Real task: interacting with GUI's
 - Pointing is fundamental



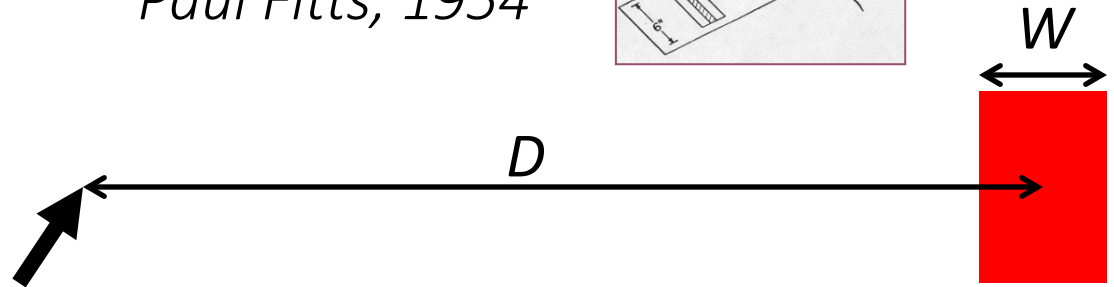
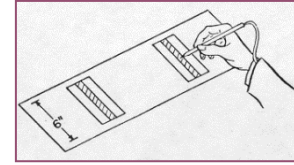
- Experimental task: target acquisition
 - abstract, elementary, essential



D is also referred to as A (amplitude) in literature.

Fitts' Law

Paul Fitts, 1954



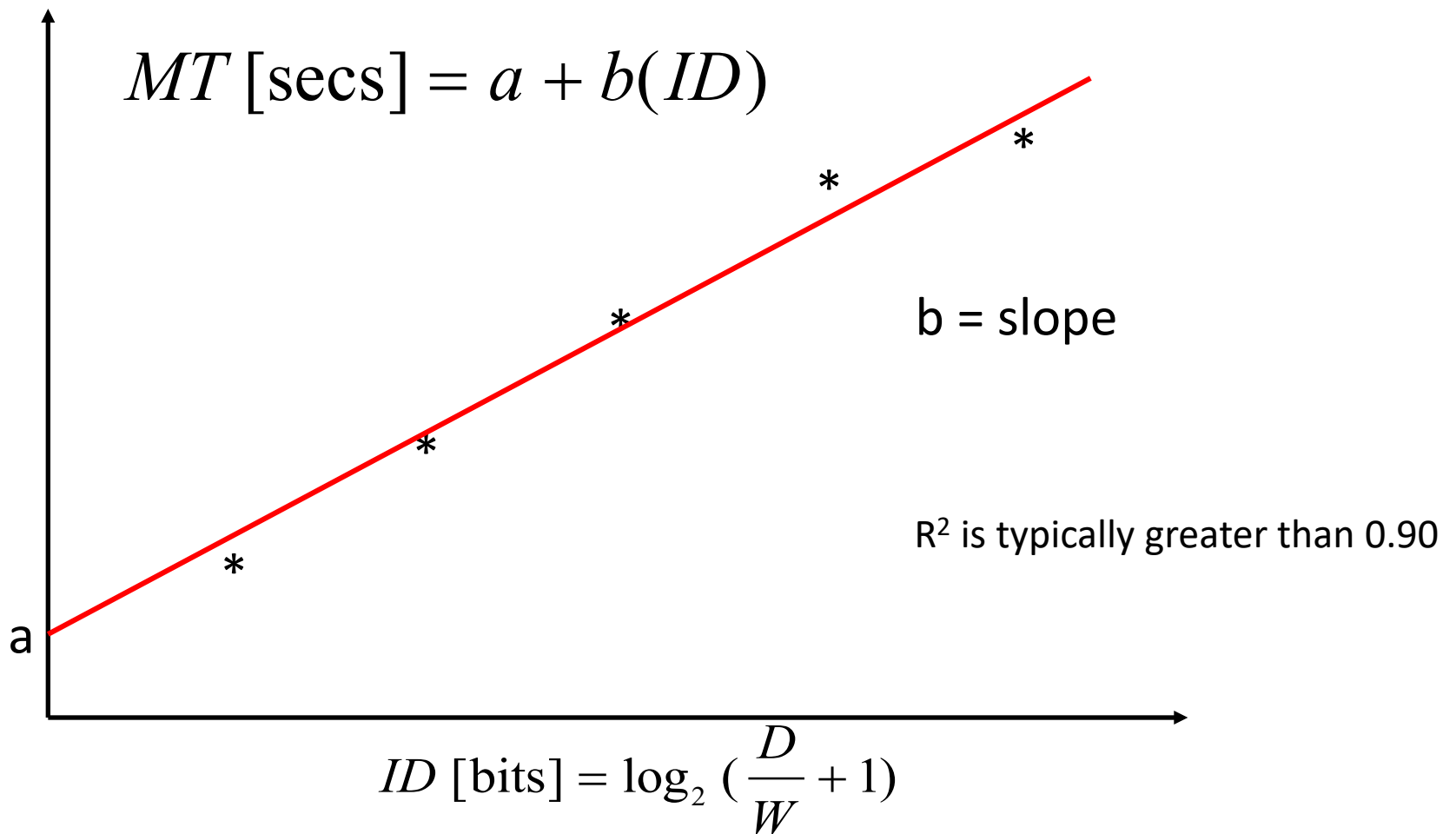
$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Movement Time

Index of Difficulty (ID [bits])

The original Fitts' law proposed by Paul Fitts in 1954:

$$MT = a + b \log_2 \left(\frac{2D}{W} \right)$$



$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Only Relative Precision (D / W) Matters

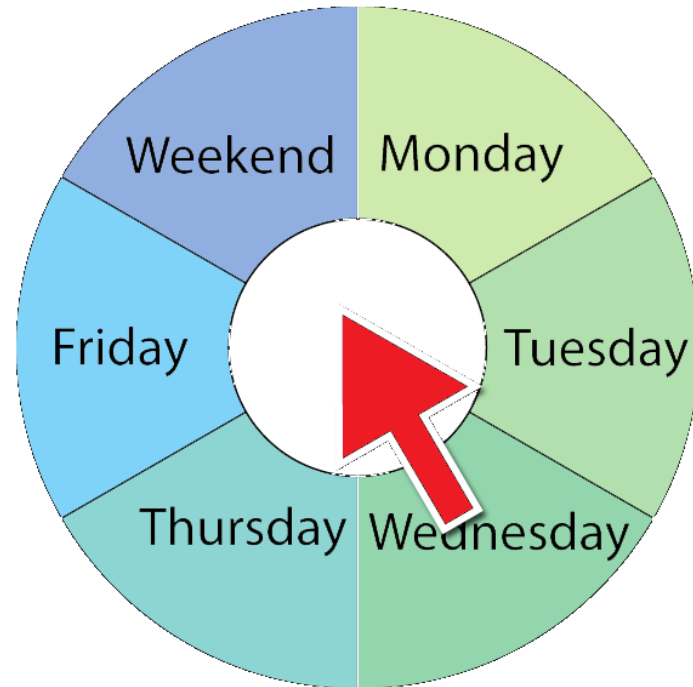
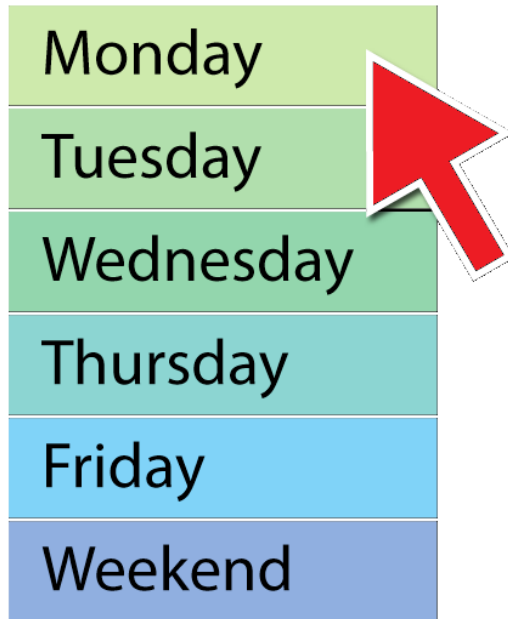
Explanation of R^2

$$R^2 \equiv 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

$$SS_{\text{res}} = \sum_i (y_i - f_i)^2 = \sum_i e_i^2$$

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2,$$

Fitts' Law Example



Which will be faster on average?

pie menu (bigger targets & less distance)

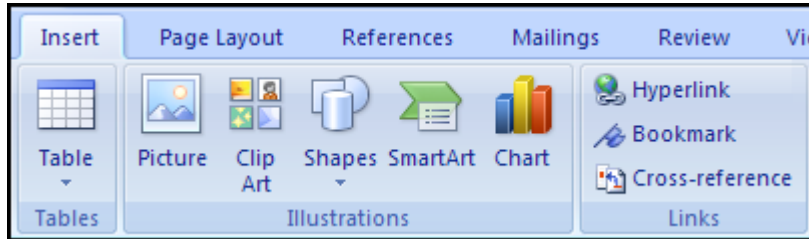
Fitts' Law in Windows and Mac OS



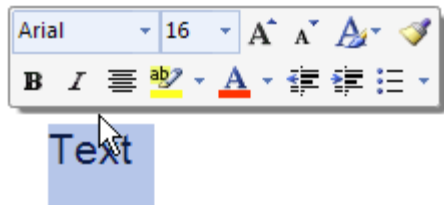
Windows 95: Missed by a pixel



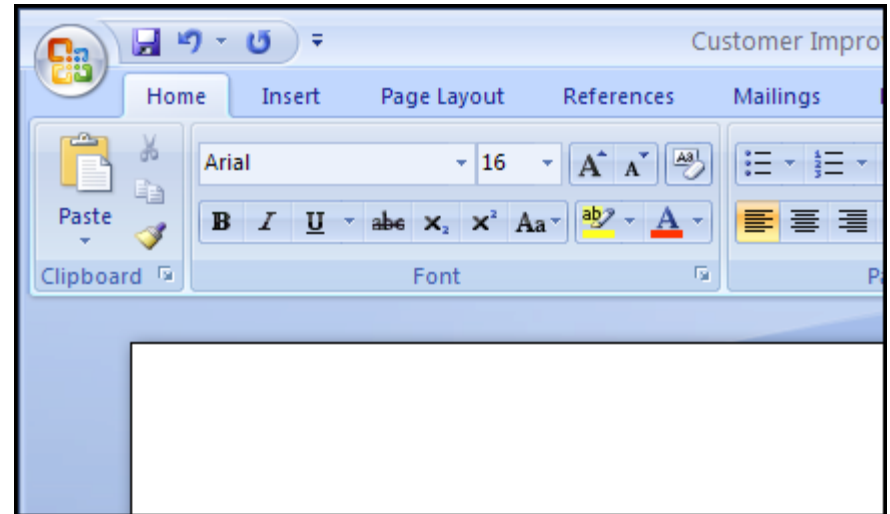
Fitts' Law in Office 2007



Larger, labeled controls can be clicked more quickly



Mini Toolbar: Close to the cursor

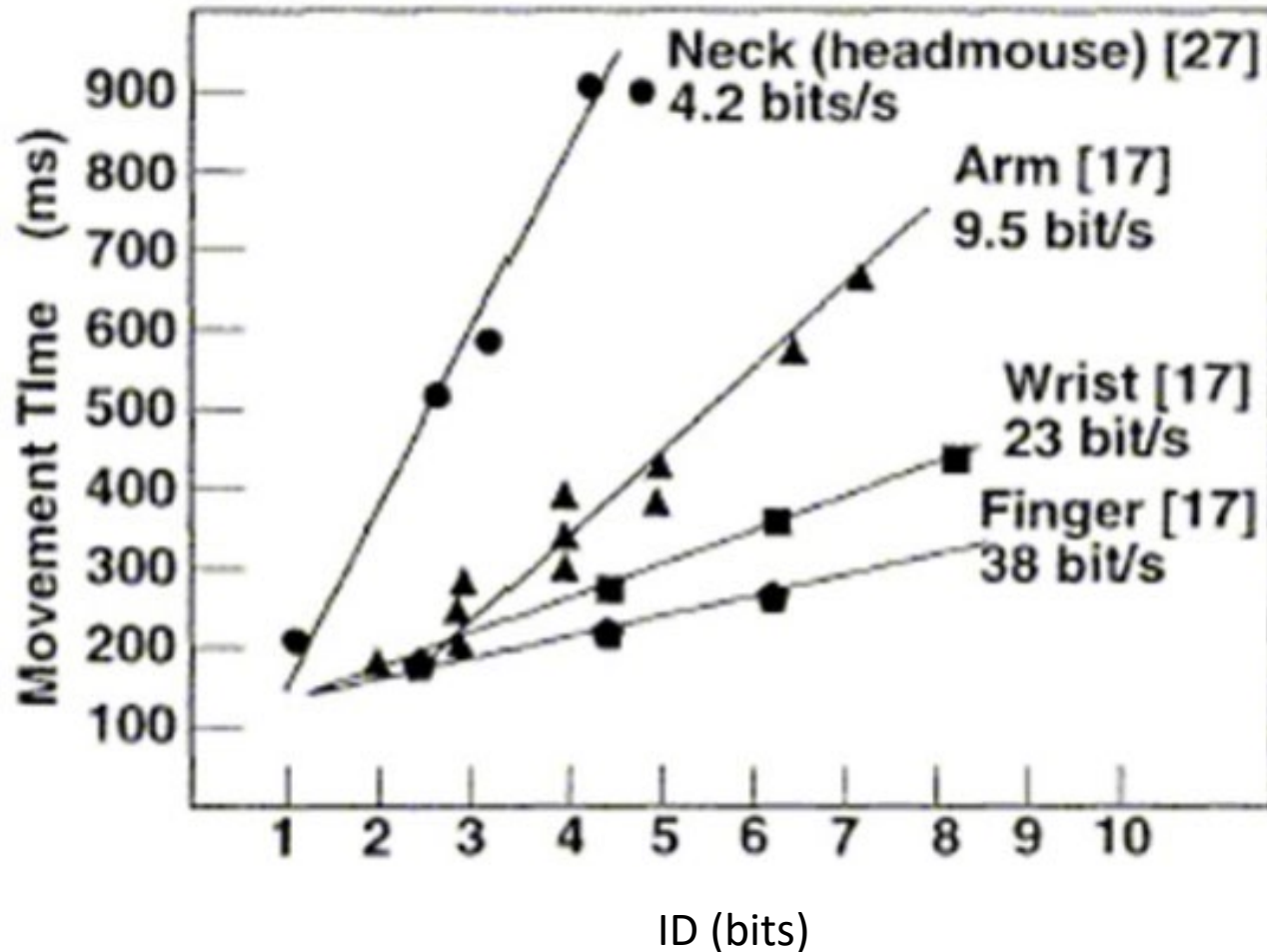


Magic Corner: Office Button in the upper-left corner

Source: Jensen Harris, An Office User Interface Blog : Giving You Fitts. Microsoft, 2007.

Bandwidth (IP) of Human Muscle Groups

$$MT = a + b \cdot ID \quad IP = 1/b$$



Which is faster?



50 years of data

Device	Study	<i>IP</i> (bits/s)
Hand	Fitts (1954)	10.6
Mouse	Card, English, & Burr (1978)	10.4
Joystick	Card, English, & Burr (1978)	5.0
Trackball	Epps (1986)	2.9
Touchpad	Epps (1986)	1.6
Eyetracker	Ware & Mikaelian (1987)	13.7

Reference:

Mackenzie, I. Fitts' Law as a research and design tool in human computer interaction. *Human Computer Interaction*, 1992, Vol. 7, pp. 91-139

Extending Fitts' law to Two-Dimensional Tasks

- Original Fitts' Law

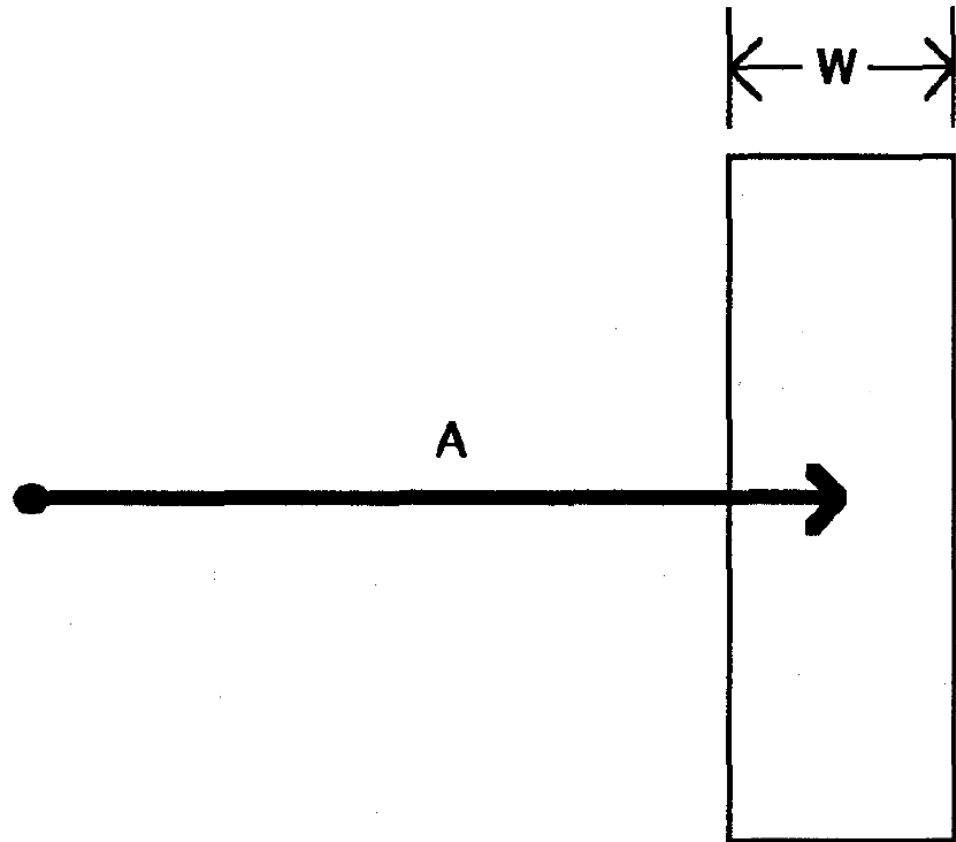
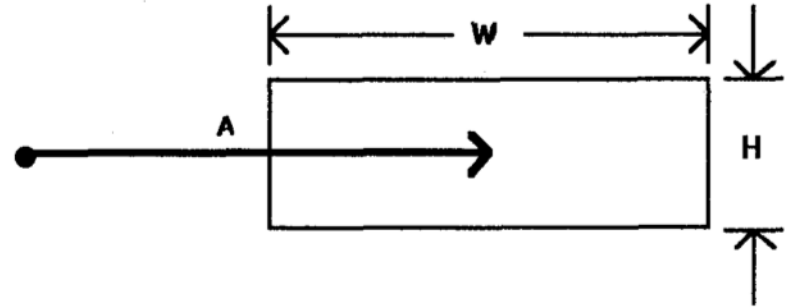


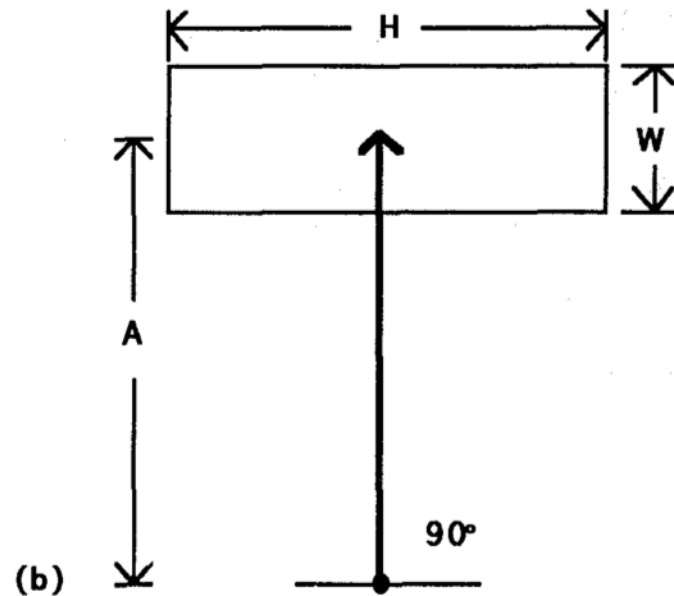
Figure 1. Fitts' law paradigm. The law is inherently one-dimensional since target amplitude (A) and width (W) are measured along the same axis.

Extending Fitts' law to Two-Dimensional Tasks

- Which one should we take into account? Width? Or Height?



(a)



(b)

Extending Fitts' law to Two-Dimensional Tasks

- How about the cases which are not horizontal or vertical?

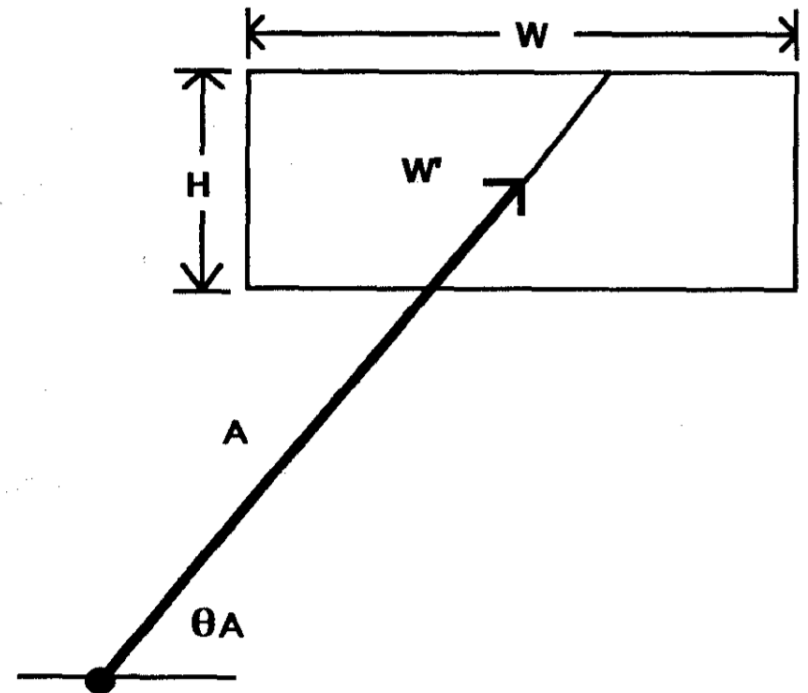


Figure 3. What is target width? Possibilities include W' (the width of the target along an approach vector) or the smaller of W or H .

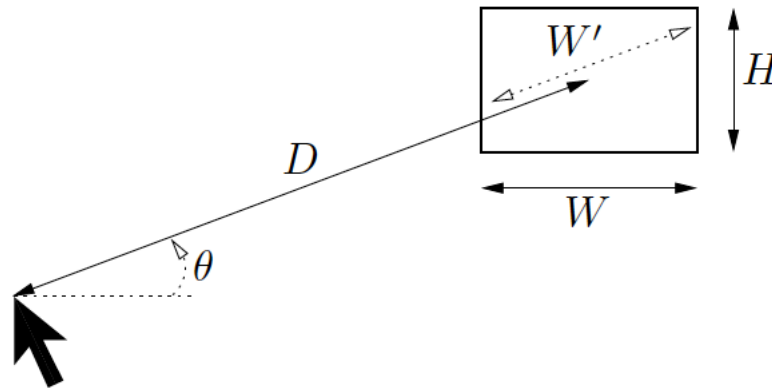
Extending Fitts' law to Two-Dimensional Tasks

- 5 model proposed
 - “STATUS QUO” model
 - “SMALLER OF” model
 - Apparent width W' model
 - Substitute W with $W*H$
 - Substitute W with $W+H$

$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

$$ID_{\min(W,H)} = \log_2 \left(\frac{D}{\min(W,H)} + 1 \right)$$

$$ID_{W'} = \log_2 \left(\frac{D}{W'} + 1 \right)$$



Extending Fitts' law to Two-Dimensional Tasks

- Experiment Result

Empirically, this is the best

Model for Target Width	<u>ID Range (bits)</u>		r^a	SE^b (ms)	<u>Regression Coefficients</u>		
	Low	High			Intercept, a (ms)	Slope, b (ms/bit)	IP (bits/s)
SMALLER-OF	1.58	5.04	.9501	64	230	166	6.0
W^1	1.00	5.04	.9333	74	337	160	6.3
$W+H$	0.74	3.54	.8755	99	402	218	4.6
$W \times H$	0.32	4.09	.8446	110	481	173	5.8
STATUS QUO	1.00	5.04	.8097	121	409	135	7.4

^a $n = 78, p < .001$

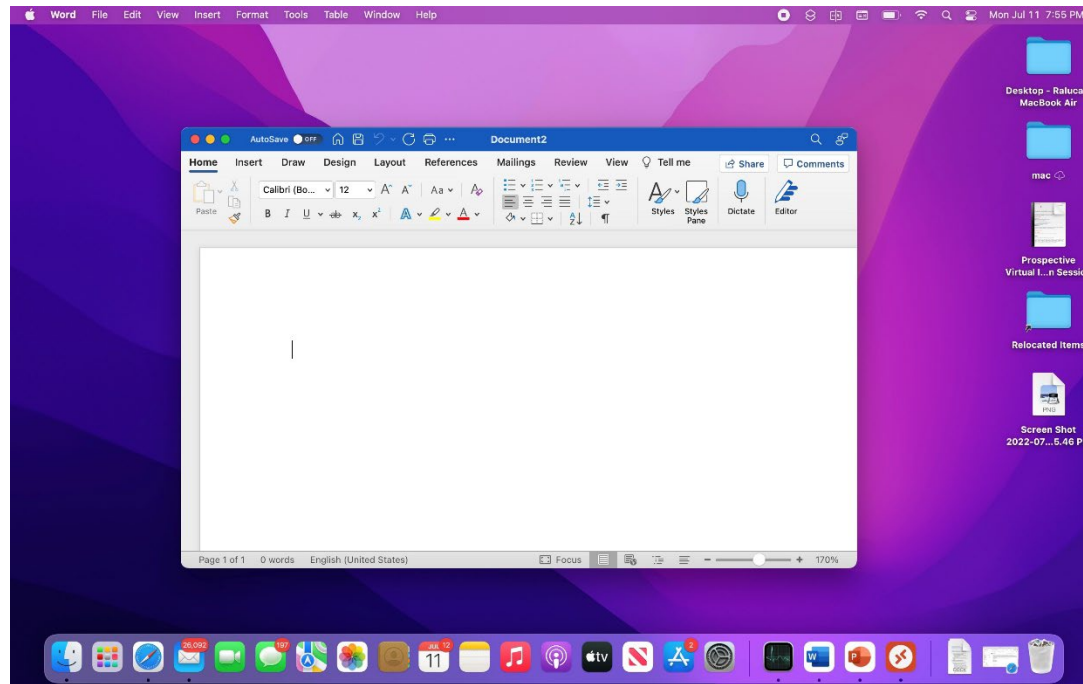
^bstandard error of estimate

Theoretically, this is similar to Fitts' Law

Figure 6. Correlations and regression coefficients for five models for target width.

Applications of Fitts' law?

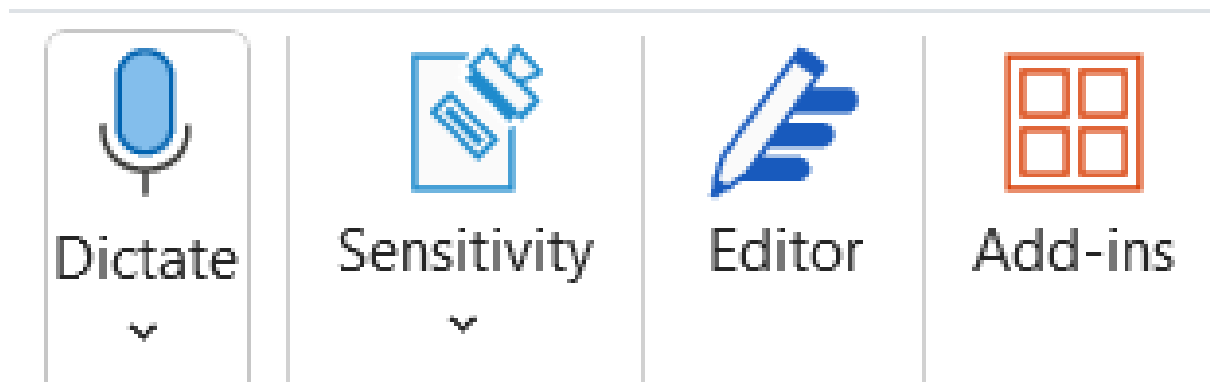
Application of Fitts' Law in UI Design



The application menu is positioned at the top edge of the screen, creating an infinite edge that allows faster access.

Application of Fitts' Law in UI Design

Icons Plus Labels



Application of Fitts' Law in UI Design

Keyboard Layout Optimization

Qwerty Layout



Q	W	E	R	T	Y	U	I	O	P
A	S	D	F	G	H	J	K	L	
	Z	X	C	V	B	N	M		

Qwerty is inefficient for one finger typing.

Optimization Objective Function

- Fitts' Law (Fitts 1954):

$$MT_{ij} = a + b \log_2 \left(\frac{D_{ij}}{W} + 1 \right)$$

MT_{ij} : Movement Time from Key i to Key j

D_{ij} : Distance from Key i to Key j

W : Key Width

Optimization Objective Function

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- Average time of typing a letter:

$$t = a + b \sum_i^{26} \sum_j^{26} P_{ij} \log_2 \left(\frac{D_{ij}}{W} + 1 \right)$$

P_{ij} : Frequency of an ordered letter pair i, j

Problem: Assuming that there is a special language called L which has three letters A, B, C only, we are designing a keyboard layout for this language. The digraph frequency of these three letters are shown in the following table. For example, the frequency of digraph BA is 0.3, and the frequency of digraph AB is 0.1. The Fitts' law parameters, which were learned from a previous study, are $a = 200$ ms, $b = 100$ (ms/bit). Please use the smaller-of 2D Fitts' law for calculation

2 nd letter 1 st letter	A	B	C
A	0.1	0.1	0.1
B	0.3	0.1	0.1
C	0.04	0.06	0.1

The keyboard layout is shown as below. Each key is a square. A user types with a pen on the soft keyboard.

A	C	B
---	---	---

1. What is the estimated movement time for typing the following digraphs, which is defined as the duration between the movement the pen leaves the first letter and moment the pen lands on the second letter? (hints: using smaller-of 2D Fitts' law)

AB, AC, BC, AA

2. What is the estimated average time of typing a letter on this keyboard layout?

Answers

1. duration for entering a digraph:

$$\text{AB: } 200 + 100 \log_2(3) = 358 \text{ ms}$$

$$\text{AC: } 200 + 100 \log_2(2) = 300 \text{ ms}$$

$$\text{BC: } 200 + 100 \log_2(2) = 300 \text{ ms}$$

$$\text{AA: } 200 \text{ ms}$$

2. What is the estimated average time of typing a letter on this keyboard layout?

$$0.1 \times 200 + 0.1 \times 358 + 0.1 \times 300 + 0.3 \times 358 + 0.1 \times 200 + 0.1 \times 300 + 0.04 \times 300 + 0.06 \times 300 + 0.1 \times 200 = 293.2 \text{ ms}$$